

I. Recap

If T is a distribution, there exists a sequence of functions

$f_1, f_2, \dots$:

$$\langle T, \chi \rangle = \int_{-\infty}^{+\infty} T(t) \chi(t) dt.$$

$$\triangleq \lim_{n \rightarrow \infty} \left[\int_{-\infty}^{+\infty} f_n(t) \chi(t) dt \right].$$

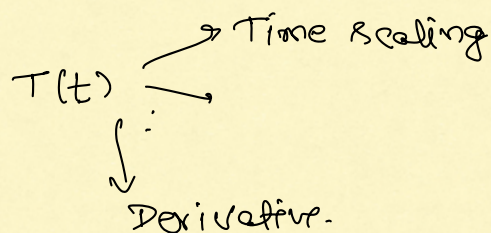
① $T = \delta$:

$$f_n(\chi) = \sqrt{\frac{n}{2\pi}} \cdot \exp(-nt^2/2).$$

② $T = 1$: $\langle T, \chi \rangle = \langle 1, \chi \rangle = \int_{-\infty}^{+\infty} \chi(t) dt$

$$f_n(t) = \exp(-t^2/n^2)$$

II. Basic Manipulations of Distributions



Assume T is a distribution associated with the sequence of

functions: $f_1(t), f_2(t), \dots$

Distribution

Function Sequence

1. $T(-t)$ $f_1(-t), f_2(-t), \dots$

$$\left[\int_{-\infty}^{+\infty} T(-t) \chi(t) dt = \lim_{n \rightarrow \infty} \int_{-\infty}^{+\infty} f_n(-t) \chi(t) dt \right]$$

2. $T(at)$ $f_1(at), f_2(at), \dots$
 $a > 0$

3. $T(t - t_0)$ $f_1(t - t_0), f_2(t - t_0), \dots$

4. $T'(t)$ $f_1'(t), f_2'(t), \dots$

5. $T(t) \times \phi(t)$ $f_1(t) \phi(t), f_2(t) \phi(t), \dots$
dist. $\downarrow$ $\downarrow$
dist. signal.

6. $T(t) * \phi(t)$ $f_1(t) * \phi(t), f_2(t) * \phi(t), \dots$

Formulae

1. $\int_{-\infty}^{+\infty} T(-t) \chi(t) dt = \int_{-\infty}^{+\infty} T(t) \chi(-t) dt$

2. $a > 0: \int_{-\infty}^{+\infty} T(at) \chi(t) dt = \frac{1}{a} \int_{-\infty}^{+\infty} T(t) \cdot \chi(t/a) dt.$

3. $\int_{-\infty}^{+\infty} T(t - t_0) \chi(t) dt = \int_{-\infty}^{+\infty} T(t) \cdot \chi(t + t_0) dt.$

$$4. \int_{-\infty}^{+\infty} T'(t) x(t) dt = - \int_{-\infty}^{+\infty} T(t) \cdot x'(t) dt$$

$$5. \int_{-\infty}^{+\infty} [T(t) \cdot \underbrace{\phi(t)}_{\text{signal}}] \cdot \underbrace{x(t)}_{\text{signal}} dt = \int_{-\infty}^{+\infty} T(t) \cdot [\underbrace{x(t) \phi(t)}_{\text{signal}}] dt.$$

$$6. \langle T * \phi, x \rangle = \langle T, \phi(-t) * x(t) \rangle$$

Proof for T' :

Assume that x is well-behaved in the following way:

$$x(t) \times \underbrace{f(t)}_{\text{Schwartz signal}} \rightarrow 0 \quad \text{as } t \rightarrow +\infty \text{ and } t \rightarrow -\infty.$$

$$\int_{-\infty}^{+\infty} T'(t) x(t) dt = \lim_{n \rightarrow \infty} \int_{-\infty}^{+\infty} \underbrace{f_n'(t) x(t) dt}_{\text{Schwartz signal}}.$$

$$\int_{-\infty}^{+\infty} f_n'(t) x(t) dt = \int_{-\infty}^{+\infty} x(t) \cdot d[f_n(t)]$$

$$= \left[x(t) \cancel{f_n(t)} \right]_{-\infty}^{+\infty}$$

$$- \int_{-\infty}^{+\infty} f_n(t) x'(t) dt$$

$$= - \int_{-\infty}^{+\infty} f_n(t) x'(t) dt.$$

$$\begin{aligned}
 \therefore \int_{-\infty}^{+\infty} T'(t) x(t) dt &= \lim_{n \rightarrow \infty} \left[- \int_{-\infty}^{+\infty} f_n(t) x'(t) dt \right] \\
 &= - \left\{ \lim_{n \rightarrow \infty} \int_{-\infty}^{+\infty} f_n(t) x'(t) dt \right\} \\
 &= - \int_{-\infty}^{+\infty} T(t) x'(t) dt.
 \end{aligned}$$

III. Examples.

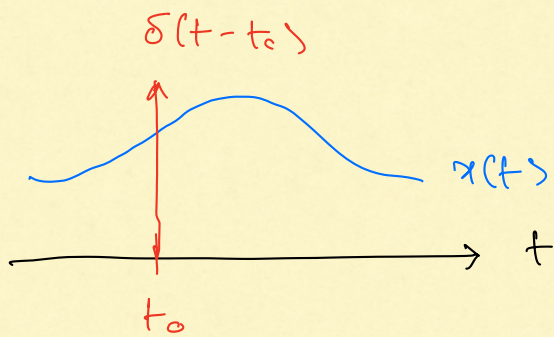
1. δ' :

$$\begin{aligned}
 \int_{-\infty}^{+\infty} \delta'(t) x(t) dt &= - \int_{-\infty}^{+\infty} \delta(t) x'(t) dt \\
 &= - x'(0) = - \frac{dx}{dt} \Big|_{t=0}
 \end{aligned}$$

$$\begin{aligned}
 \delta'': \quad \int_{-\infty}^{+\infty} \delta''(t) x(t) dt &= - \int_{-\infty}^{+\infty} \delta'(t) x'(t) dt \\
 &= - \left[-(x')'(0) \right] = x''(0).
 \end{aligned}$$

2. $\delta(t - t_0)$:

$$\begin{aligned}
 \int_{-\infty}^{+\infty} \delta(t - t_0) x(t) dt &= \int_{-\infty}^{+\infty} \delta(t) \cdot x(t + t_0) dt \\
 &= x(t_0)
 \end{aligned}$$



$$\begin{aligned}
 \textcircled{8} \quad \delta(-t): \quad & \int_{-\infty}^{+\infty} \delta(-t) x(t) dt = \int_{-\infty}^{+\infty} \delta(t) x(-t) dt \\
 & = x(0) \\
 & = \int_{-\infty}^{+\infty} \delta(t) x(t) dt
 \end{aligned}$$

Thus, $\delta(t) \succ \delta(-t)$ act the same way on all x .

$$\Rightarrow \delta(t) = \delta(-t)$$

$\Rightarrow \delta$ is an even distribution.

$$\begin{aligned}
 4. \quad \int_{-\infty}^{+\infty} \delta(t) \underbrace{\phi(t)}_{\text{Signal}} x(t) dt &= \int_{-\infty}^{+\infty} \delta(t) [\phi(t) x(t)] dt \\
 &= \phi(0) x(0) \\
 &= \phi(0) \cdot \int_{-\infty}^{+\infty} \delta(t) x(t) dt \\
 &= \int_{-\infty}^{+\infty} [\phi(0) \cdot \delta(t)] x(t) dt.
 \end{aligned}$$

$$\Rightarrow \delta(t) \cdot \phi(t) = \phi(0) \cdot \delta(t)$$

5. What is the first-derivative of $u(t)$?
(as a distribution)

$$\begin{aligned} \int_{-\infty}^{+\infty} \underbrace{u'(t)}_{\text{distribution}} \gamma(t) dt &= - \int_{-\infty}^{+\infty} u(t) \cdot \gamma'(t) dt \\ &= - \int_0^{\infty} \left[\int_0^{\infty} \gamma'(t) dt \right] \\ &= - \left[\gamma(t) \right]_0^{\infty} \end{aligned}$$

$$= - [\gamma(\infty) - \gamma(0)] = \gamma(0)$$

if γ is Schwarz: $\gamma(\infty) = 0$

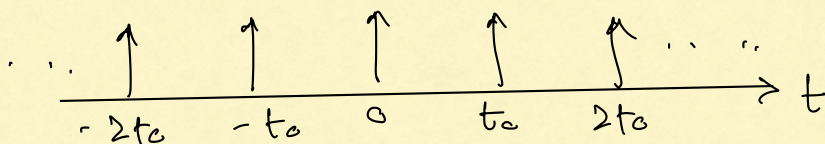
$$= \int_{-\infty}^{+\infty} \delta(t) \gamma(t) dt \quad \forall \gamma.$$

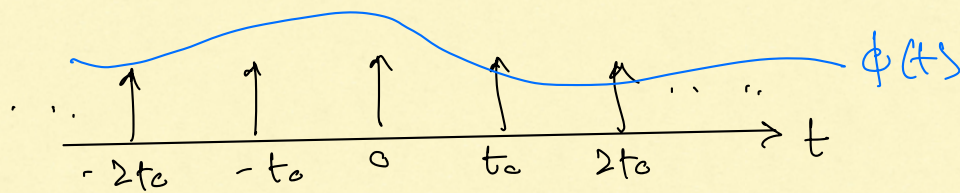
$$\therefore u'(t) = \delta(t).$$

6. Homework: $\text{sgn}' = 2\delta$

7. Sampling:

$$T = \sum_{k=-\infty}^{+\infty} \delta(t - k t_0) \leftarrow \begin{array}{l} \text{impulse train} \\ \text{or} \\ \text{Dirac comb.} \end{array}$$





$$\int_{-\infty}^{+\infty} \left[\phi(t) \cdot \sum_k \delta(t - kt_0) \right] \chi(t) dt$$

$$= \int_{-\infty}^{+\infty} \left[\sum_k \delta(t - kt_0) \right] \cdot [\phi(t) \chi(t)] dt$$

$$= \sum_k \phi(kt_0) \cdot \chi(kt_0)$$

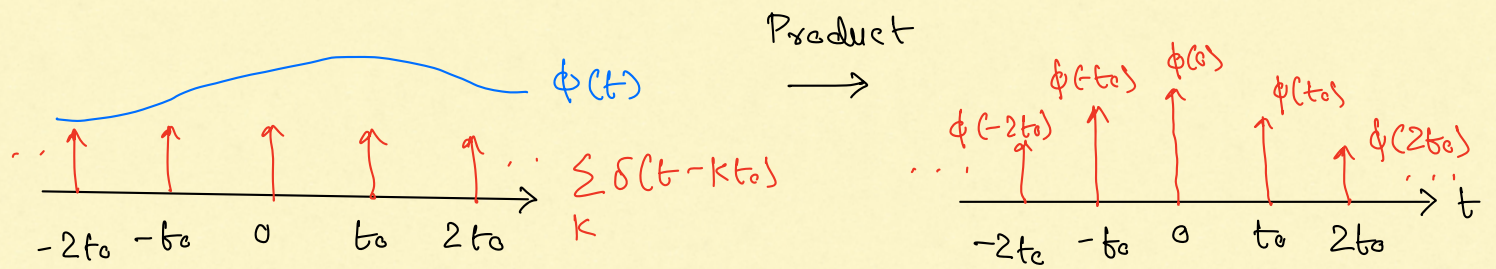
$$= \sum_k \phi(kt_0) \cdot \int_{-\infty}^{+\infty} \delta(t - kt_0) \chi(t) dt$$

$$= \int_{-\infty}^{+\infty} \left[\sum_k \phi(kt_0) \cdot \delta(t - kt_0) \right] \chi(t) dt.$$

∴ As distributions:

$$\underbrace{\sum_k \delta(t - kt_0)}_{\text{Dirac comb}} \times \underbrace{\phi(t)}_{\text{Signal}} = \sum_k \underbrace{\phi(kt_0)}_{\text{Scalar}} \cdot \underbrace{\delta(t - kt_0)}_{\text{shifted delta}}$$

↓
↓
 Product
 ↓
Sum of scaled & shifted deltas.



Note:

This new distribution is a mathematical record of all the values of $\phi(t)$ at $t = kt_0$,

$k \in \mathbb{Z}$:

$$\sum_{k=-\infty}^{+\infty} \underbrace{\phi(kt_0)}_{\text{value of } \phi \text{ at this time instance}} \cdot \underbrace{\delta(t - kt_0)}_{\text{time instance}}$$

value of ϕ at this time instance.

Multiplying $\phi(t)$ with $\sum_k \delta(t - kt_0)$ "samples" the signal $\phi(t)$ at all $t = kt_0$ but loses the information about $\phi(t)$ at all other instances $t \neq kt_0$, $k \in \mathbb{Z}$.

In signal processing

signal $\leftarrow \{ \phi(t) \rightarrow \phi(kt_0), k \in \mathbb{Z} \} \rightarrow \text{samples.}$

this map is called sampling

Mathematically, sampling is done by $\phi(t) \rightarrow \phi(t) \cdot \sum_k \delta(t - kt_0)$

In real-life, sampling (and quantization) is performed by

analog-to-digital converters (ADC).